Google Cloud H2S

AI Builder Cup

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Build for the world.



Team Details

a. Team name: **Xcore**
b. Team leader name: **Thanveer Ahamed**
c. Problem Statement: Intelligent Risk, Fraud & Financial Experiences (BFSI). Lenders and insurers judge property risk from slow site visits or self-reported photos nobody can verify. We turn a 20-second phone video into a fraud-checked, explainable, underwriting-ready property risk file.

Property Inspection AI

Phone video → underwriting-ready property risk file



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Brief about the idea

From a 20-second phone video to an underwriting-ready property risk file

Property Inspection AI is a Gen AI inspection service for mortgage lenders, insurers and inspectors. Upload a phone walkthrough video; in minutes it returns defects, condition grade A–E, perils, repair cost, valuation impact, premium loading, an underwriting decision with reasons, fraud flags **and a 3D model that proves what was inspected.**

THE PROBLEM: property risk is judged from photos nobody can verify

Slow site visits	Unverifiable photos	No proof of coverage	Subjective grading	Overvaluation
Days of waiting; remote cases go unchecked	Reused, edited, off-screen or wrong property	Which walls did the inspection see?	Inspectors disagree; reasons unrecorded	RBI has flagged inflated collateral values

WHO IT SERVES: one inspection record, five BFSI decision-makers

Lenders, HFCs, NBFCs	Insurers: underwriting	Insurers: claims	Inspectors & valuers	Owners & buyers
Collateral grade, valuation impact, REFER flag	Perils, risk level, premium loading	Repair-cost range, fraud-checked evidence	AI-drafted report, re-film list	Condition report, 3D model, floor plan

$308.6B
yearly US insurance fraud, all lines (Coalition Against Insurance Fraud, 2022)

~$45B
of that in property & casualty insurance (same study)

$4.5–12.5B
global home-inspection services market, 2025 (industry reports)



Minutes, not days · Fraud-resistant · Explainable

Concept illustration of the product

Opportunities

How it is different, how it solves the problem, and our USP

HOW DIFFERENT IS IT FROM EXISTING IDEAS?

	Site visit / photo-only AI tools	Property Inspection AI
Turnaround	Days (visit) / minutes (tools)	**Minutes; ~10 min per 20 s video**
Proof of what was inspected	Inspector's word / none	**10 cm coverage map + 3D-pinned findings**
Evidence fraud checks	None / limited	**Keyframe dHash reuse, Gemini screen / AI check**
How the decision is made	Subjective / black-box score	**Transparent YAML rules; Gemini explains**
False-positive control	Human / single model	**Find (LocateAnything) → verify (Gemma 4) → rules**
BFSI outputs & integration	Report or damage list, PDF	**Grade, perils, cost, valuation, premium; JSON via REST API**

HOW WILL IT SOLVE THE PROBLEM?

Slow site visits →	full risk file in minutes; low-risk homes clear without a visit
Unverifiable photos →	every walkthrough fraud-checked; any HIGH flag → REFER
No proof of coverage →	< 60% AI-inspected surface → REFER with a re-film list
Subjective grading →	score = 100 − Σ(points × severity × state weight)
Overvaluation →	valuation-impact %, repair retention, REFER before disbursal

USP of the proposed solution

Proof of inspection. A 10 cm coverage map and 3D-pinned findings show exactly what the AI saw, and what to film again.

Find → verify → decide. Two models cut false positives; deterministic rules make every decision auditable.

Fraud-resistant evidence. Recycled videos, screen recordings and AI footage are flagged before a decision.

One record, five teams. Lending, underwriting, claims, inspectors and owners share one JSON record behind a REST API.

Under $1 per video. ≈ $0.50 of L4 time + ≈ $0.35 of Gemini per walkthrough; the GPU Job runs only per video.

List of features offered by the solution

One upload gives a complete, explainable risk file

1. **Defects localised & verified**
   18 defect types on 12 components
2. **Condition grade A–E**
   0–100 score, transparent formula
3. **Risk level & insurance perils**
   Water, structural, moisture...
4. **Indicative repair-cost range**
   Per finding and per property
5. **Valuation & collateral impact**
   Impact % and repair retention
6. **Premium-loading suggestion**
   For quote and renewal
7. **Underwriting decision + reasons**
   PROCEED / CONDITIONS / REFER
8. **Evidence-integrity flags**
   Recycled video, screens, AI footage
9. **Video → 3D, floor plan, coverage**
   GLB models, room areas, maps
10. **Gemini analysis & Q&A**
    Grounded answers citing issue IDs

✓ Detection scope

Walls, ceilings, floor & wall tiles, grout, wood/vinyl, countertops, doors, locks, windows, trim, kitchen exhausts in 8 room types. Look-alikes (grout, grain, shadows) rejected; locks, windows, fans always manual; cracks get a structural note, water/mold a moisture check.

✓ Fraud & evidence integrity

Keyframe dHash reuse across every property, Gemini screen / render / AI check on every keyframe, blurred frames dropped, video coverage < 60% → REFER. Any HIGH flag → human review.

✓ 3D digital twin & coverage map

Depth Anything 3 + OpenCV 5 SLAM + Open3D TSDF → clean & SLAM GLB, floor plan (PNG + SVG), navigation map. Every 10 cm of surface checked; greedy AI keyframes; findings pinned to a wall and video time.

✓ Gemini co-pilot & product

Property analysis, grounded Q&A, scene and object naming (explains, never changes results). Portfolio dashboard, live 3D-scan view, Lite / Advanced 3D, 3D viewer, HTML / Markdown / JSON reports, REST API at /docs.

Process flow diagram or Use-case diagram

Capture, inspect, decide, act: minutes instead of days

ACTORS

- **Inspector / owner** captures
- **Underwriter / claims** reviews & decides
- **Core systems** LOS, policy, claims

1 Capture
A phone walkthrough video. No special hardware.

→

2 AI inspection
Gemini reads the room; LocateAnything finds defects; Gemma 4 verifies each crop.

→

3 Risk decision
Rules set grade, perils, cost, valuation, premium, decision; Gemini explains.

→

4 Act
Report, Gemini Q&A and a JSON record via the REST API to core systems.

DECISION OUTCOMES (back to underwriter and core systems)

- PROCEED
- PROCEED W/ CONDITIONS: note defects
- PROCEED W/ CONDITIONS: repairs / retention
- REFER: professional survey
- REFER: evidence integrity
- REFER: additional evidence

KEYFRAME PIPELINE

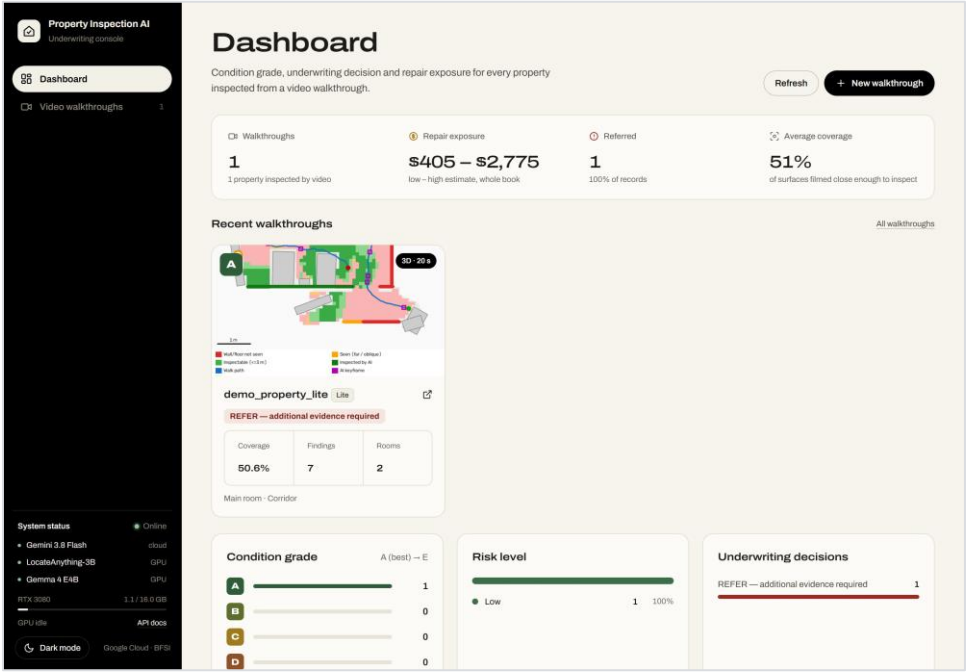
1 Quality + integrity (dHash) → 2 Gemini: room, components, authenticity → 3 Prompt router: 18 defects, 12 components → 4 LocateAnything-3B boxes → NMS → 5 Full-res crops → Gemma 4 verify → 6 Rule engine → status → risk engine

VIDEO PIPELINE

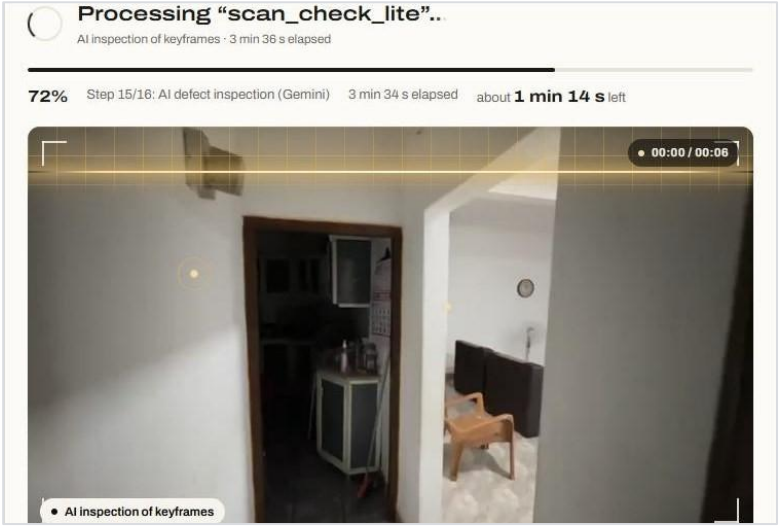
1 Depth Anything 3 depth per frame → 2 OpenCV 5 SLAM camera poses → 3 Open3D TSDF surface fusion → 4 Walls, floor, doors; Gemini names objects → 5 10 cm coverage + AI keyframes → 6 Keyframes → keyframe pipeline; pins in 3D

Wireframes/Mock diagrams of the proposed solution (optional)

Actual product screens: an inspection app built for underwriters



Dashboard: walkthroughs, repair exposure, referrals, coverage and condition grades across the book

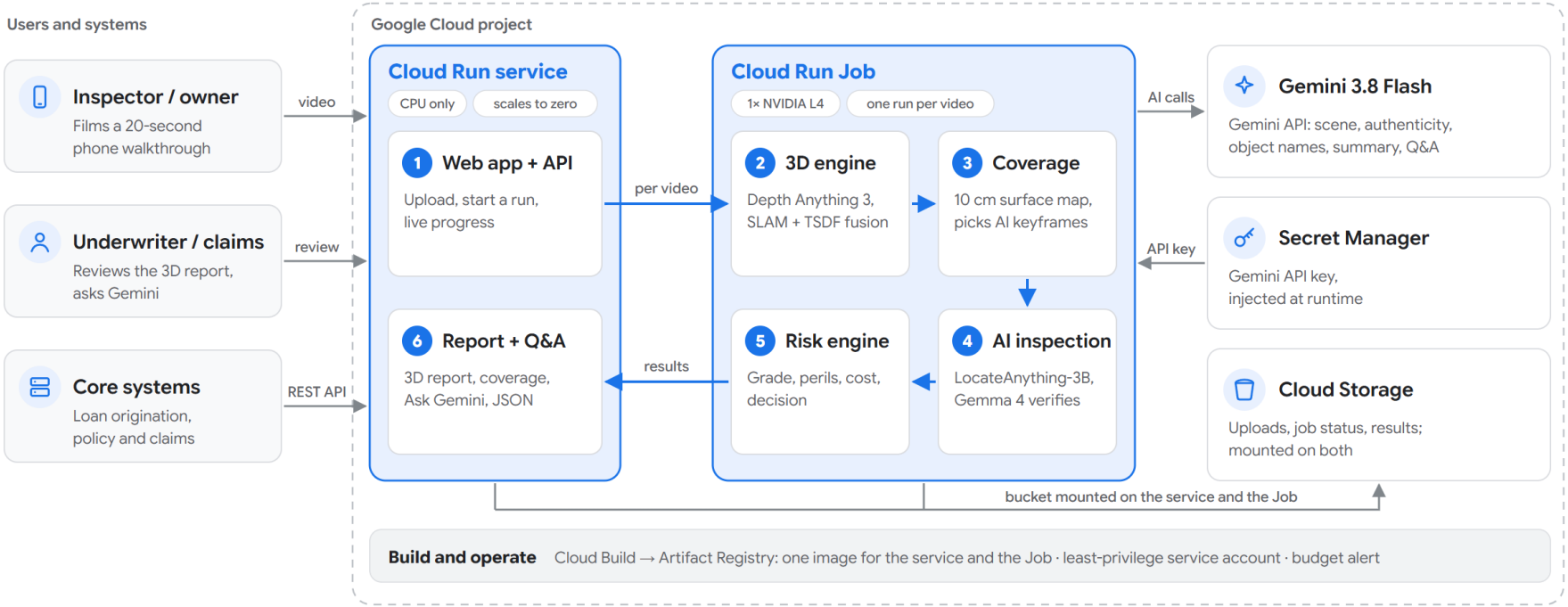


Video walkthrough: live 3D-scan progress while the video is processed

- Dashboard → Video walkthroughs → 3D report → Ask Gemini
- Video upload with a live 3D-scan view, Lite or Advanced
- Interactive 3D viewer: walk inside, see coverage gaps
- Reports in HTML, Markdown, JSON; REST API at /docs

Architecture diagram of the proposed solution

Cloud Run web service + L4 GPU Job, powered by Gemini



Deployed layout (v1.0.0): web UI + API on a CPU-only Cloud Run service; 3D, coverage and AI inspection on a Cloud Run Job with one NVIDIA L4 per video; Cloud Storage bucket mounted on both; Gemini 3.8 Flash via the Gemini API (key in Secret Manager). BigQuery and Pub/Sub are not used.

Technologies to be used in the solution

Built on Google AI and Google Cloud, with best-in-class open models

GOOGLE AI & CLOUD, IN THE PROTOTYPE

- ✓ **Gemini 3.8 Flash** — Gemini API: scene, authenticity, analysis, grounded Q&A, 3D object names
- ✓ **Gemma 4 (E4B)** — Open Google model that verifies every defect crop on the GPU
- ✓ **Cloud Run + L4 GPU Job** — CPU web service + one L4 Job per video; both scale to zero
- ✓ **Cloud Storage** — Bucket mounted on both: uploads, job status, results
- ✓ **Secret Manager** — Gemini API key injected at runtime; none in the image
- ✓ **Cloud Build** — Builds the ~12 GB image; weights fetched inside the build
- ✓ **Artifact Registry** — One image with baked-in models for the service and the Job
- ✓ **IAM service account + Logging** — Least-privilege runtime account; no keys in files

OPEN MODELS & LIBRARIES

- **LocateAnything-3B** open-vocabulary defect boxes
- **llama.cpp** local GPU inference for Gemma 4
- **Depth Anything 3** monocular depth per frame
- **OpenCV 5** SLAM tracking, ICP, CV evidence
- **Open3D** TSDF fusion, meshes, GLB export
- **gsplat** Gaussian-splatting experiment
- **FastAPI + Python** 18 REST endpoints, web UI, /docs
- **YAML rule engine** client-owned risk rules

NEXT ON THE ROADMAP

- **Document AI** — Title deeds, valuation reports, repair invoices cross-checked with images
- **Vertex AI Search** — Policy wording and lending guidelines for the assistant
- **Vertex AI + BigQuery ML** — Claim-likelihood and collateral-risk models
- **Cloud SQL / AlloyDB** — Multi-tenant case management, audit trail, SSO

Why this stack: Gemini does reasoning, authenticity and explanation; open models on the Cloud Run GPU Job do high-volume boxing and verification cheaply; deterministic rules keep decisions auditable. Data stays in the client's own Google Cloud project, and the JSON record and REST API plug straight into lending, underwriting and claims systems.

Estimated implementation cost (optional)

Priced per inspection, far below a physical site visit

UNIT ECONOMICS (INDICATIVE, FROM THE PROTOTYPE)

Cost item	Basis	Estimate
Gemini calls (Gemini API)	≈ $0.005 per call	**≈ $0.35 per walkthrough**
NVIDIA L4 on a Cloud Run Job	1× L4, 8 vCPU, 32 GiB; one run per video	**≈ $0.50 per 20-min walkthrough**
Video inspection, all-in	L4 Job + Gemini calls	**≈ $0.85 per walkthrough**
Web UI + API (Cloud Run)	1 vCPU, 2 GiB, scales to zero	**Within the free tier**
Idle footprint	12 GB image + data bucket	**≈ $1.20 per month**
Deployment & ops	One-command deploy, $30/mo budget alert	**No servers to manage**

≈ $0.35
Gemini per walkthrough

≈ $0.50
L4 GPU time per video

$0
GPU cost when idle (Job runs per video)

BUSINESS MODEL

Pay-per-inspection API
Lenders and insurers send a walkthrough video from their own apps; get JSON, a report and a decision back.

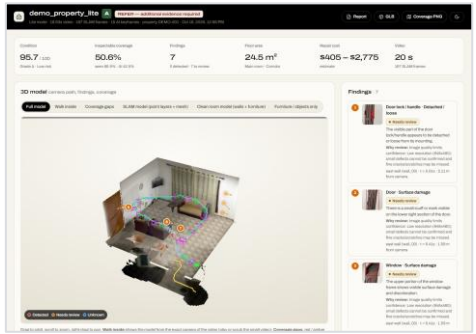
Enterprise subscription
Portfolio dashboard, BigQuery export, custom risk rules, SSO, private deployment in the client's GCP project.

Inspector & valuer firms
White-label app: AI drafts every report and coverage control, so each inspector handles more cases.

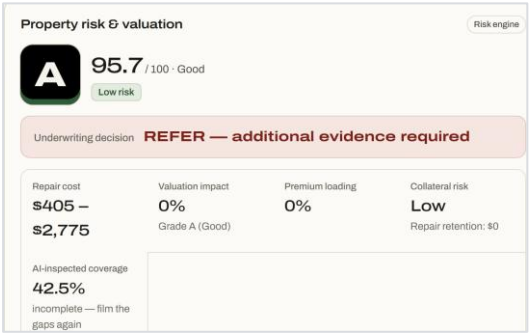
Pricing hypothesis to validate in pilots: a per-inspection fee well below a physical visit, with volume tiers for portfolios.

Snapshots of the prototype

Real screens and outputs from the working prototype v1.0.0



3D report: findings, grade A, decision



Risk & valuation panel: REFER with reasons



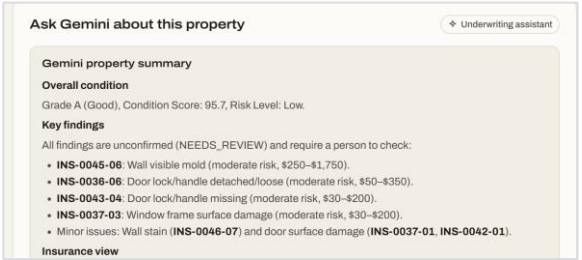
Keyframe detection: door and window damage



Phone video → clean 3D model (GLB)



Coverage map: what to film again



Gemini property analysis (co-pilot)

Prototype Performance report/Benchmarking

Measured on a real 20-second walkthrough, plus a fair 4-pipeline evaluation

CASE STUDY: REAL HOME WALKTHROUGH (MEASURED)

19.6 s phone video	**197** SLAM frames	**15** AI keyframes
24.5 m² 2 rooms measured	**95.7 · A** condition score	**$405–2,775** indicative repairs

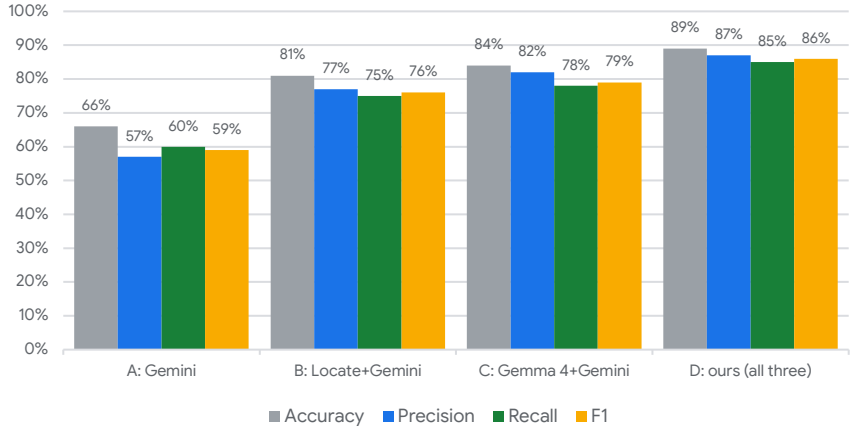
Decision: REFER, additional evidence required

Only 42.5% of surfaces were AI-inspected (min 60%), so the app lists the walls and ceilings to film again. 7 possible defects pinned in 3D (Needs review).

SPEED & COST (FROM THE PROTOTYPE)

10.6 min end to end, 20 s video, laptop GPU	**11.1 / 26.9 min** 3D build, Lite / Advanced
21.8 / 21.6 dB PSNR, Lite / Advanced	**≈ $0.35** Gemini per walkthrough

EVALUATION: 4 PIPELINES · 10 ROOMS · 100 REGIONS



Simulated illustration, not yet measured: values show the report format and the expected gain; they will be replaced after running the protocol on D.

Fair test: same images, schema, Gemini settings and decision rule; test set frozen; 5 living rooms + 5 bedrooms incl. 2 clean controls; 40 defective / 60 clean regions.

Targets for D: ≥ 85% accuracy, precision, recall, F1; ≤ 1 false alarm on clean rooms; beat A on every metric.

Additional Details/Future Development (if any)

Roadmap: an inspection agent with spatial reasoning, then portfolio risk

NOW · v1.0.0

- Video walkthrough inspection
- Risk and valuation engine
- Fraud and coverage checks
- Gemini analysis and Q&A
- Cloud Run service + GPU Job

NEXT · 3 months

- Inspection agent: Gemini + Agent Platform (ADK)
- Live capture guidance: what to film next
- Spatial reasoning over the 3D model
- Answers for lender, insurer, surveyor, owner

LATER · 6–12 months

- BigQuery ML claim & collateral-risk scoring
- Before/after claim comparison in 3D
- Document AI + Vertex AI Search: deeds, policies
- Agent hand-offs to core systems, human approval

WHAT WE BUILT: FROM DEPTH RESEARCH TO A DEPLOYABLE PRODUCT

01 Depth + SLAM
Depth Anything 3 visual SLAM, occupancy grid, A* path

02 Hybrid SLAM
OpenCV 5 tracking, ICP, Manhattan planes

03 Photo-real 3D
Gaussian Splatting experiment (gsplat)

04 Production 3D
Video → GLB models, floor plan (+SVG)

05 Inspection AI
Taxonomy, locate + verify, risk engine, fraud

06 Google Cloud v1.0.0
Gemini API, Cloud Run service + L4 Job

18
defect types, 12 components

8
room types

18
REST API endpoints

2
3D modes (Lite / Advanced)

1
command to deploy on GCP

Additional Details/Future Development (contd.)

Responsible AI and the go-to-market plan

RESPONSIBLE AI: HUMANS DECIDE, AI SHOWS EVIDENCE

1 Human in the loop

Decision support only; a qualified surveyor or underwriter makes the final call.

2 Explainable

Deterministic YAML rules; every decision lists its reasons and risk points.

3 Auditable

Raw outputs archived; each inspection a stored JSON record; versioned config.

4 Safe wording

No structural-safety or root-cause claims; locks, fans, windows stay manual.

5 Secure

Secret Manager for the Gemini key; data stays in the client's project.

6 Licence-aware

LocateAnything is non-commercial; Gemini localisation is the drop-in swap.

GO-TO-MARKET: HOUSING FINANCE & HOME INSURANCE, APAC

PILOT · 0–3 months

1 housing-finance lender + 1 home insurer; AI alongside existing inspections on ~500 cases.

INTEGRATE · 3–6 months

Pub/Sub into loan-origination and claims; BigQuery dashboards; custom rules per client.

SCALE · 6–18 months

Google Cloud Marketplace, valuer & inspector networks, commercial property and claims.

Pilot KPIs: turnaround · cost per inspection · % auto-cleared vs referred · fraud flags confirmed · re-inspection rate

"Every lender and insurer already asks for photos. We replace them with a short phone video that is trustworthy, explainable and ready for a decision."

Provide links to your:

- GitHub Public Repository: github.com/XCore-Lab/property-inspection-ai
- Demo Video Link (3 Minutes): [paste link here]
- Final Product Link: property-inspection-ai-77055350047.us-central1.run.app



Property Inspection AI · Prototype v1.0.0 · one-command Cloud Run deploy

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

